

# Atlas Bill of Materials

Atlas is a refurbished HP Z840 workstation used as a two-GPU ML and LLM host. Everything below was read off the running machine on 2026-09-16 (dmidecode, lscpu, lspci, lsblk, mdadm, nvidia-smi) unless marked “owner-stated” or “not readable from software”.

## Platform

| Qty | Item                    | Part / model                                                                                  | Notes                                                                   |
|-----|-------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1   | Chassis and motherboard | HP Z840 Workstation, system board HP 2129, BIOS M60 v02.34 (2017-05-18)                       | Dual-socket LGA2011-3, C612 chipset, 7 PCIe slots, 4 internal 3.5” bays |
| 1   | Power supply            | HP 1125 W (owner-stated)                                                                      | Required for two 250 W GPUs plus two 135 W CPUs                         |
| 2   | CPU                     | Intel Xeon E5-2690 v3, 12 cores / 24 threads, 2.6 GHz base, 3.5 GHz turbo, 135 W TDP          | 48 threads total                                                        |
| 2   | CPU cooler              | HP Z840 liquid cooling heatsink module (owner-stated), HP part family 749599-001 / 635869-00x | Still reaches 99 °C at full load; cap jobs at about 20 threads          |
| 16  | Memory                  | SK Hynix HMA42GR7MFR4N-TF, 16 GB DDR4-2133 ECC RDIMM                                          | 256 GB total, all 16 slots filled, 8 per CPU                            |

## GPUs

| Qty | Item | Part / model                                                                                           | Notes                                                                                                                                                                                               |
|-----|------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | GPU  | NVIDIA Quadro GV100, 32 GB HBM2, Volta GV100GL, 5120 CUDA cores, 640 tensor cores, 250 W, PCIe 3.0 x16 | One per CPU (bus 04:00 on CPU0, 84:00 on CPU1). NVLink is not possible on the Z840 because each CPU exposes one x16 root port, so the bridge cannot span sockets. GPU-to-GPU traffic goes over QPI. |

## Storage

| Qty | Item     | Part / model                                                                       | Role                                |
|-----|----------|------------------------------------------------------------------------------------|-------------------------------------|
| 1   | NVMe SSD | Samsung 970 EVO Plus 2 TB (M.2 on a PCIe adapter card, adapter model not readable) | OS root, Ubuntu LVM, 1.8 TB         |
| 2   | SATA SSD | Samsung 870 series 1 TB                                                            | md1 RAID1 mirror, 1 TB              |
| 3   | SAS HDD  | HGST Ultrastar He8 HUH728080AL4200, 8 TB, 7200 rpm, SAS 12 Gb/s, 4Kn               | md5 RAID5, 16 TB usable, mounted at |

| Qty | Item             | Part / model                                                              | Role                                                                      |
|-----|------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 2   | SATA SSD (spare) | Samsung PM853T MZ7GE960, 960 GB enterprise                                | /archive<br>Not in any active array (one carries a stale RAID superblock) |
| 1   | HBA              | HP-branded LSI SAS2308 8-port SAS-2 HBA (HP subsystem 103c:158b, mpt3sas) | Drives the SAS HDDs and two of the SSDs                                   |
| 1   | Optical          | HL DVD-RAM GUB0N slim                                                     | Unused                                                                    |

Onboard Intel C612 SATA controllers (two, in RAID mode) carry the other SSDs.

## Networking and peripherals

| Qty | Item               | Part / model                                                 | Notes        |
|-----|--------------------|--------------------------------------------------------------|--------------|
| 2   | Ethernet (onboard) | Intel I218-LM (eno1, in use) and Intel I210 (enp5s0, unused) | Both gigabit |
| 1   | Keyboard           | Logic3 G-720                                                 |              |
| 1   | Mouse              | Logitech M90/M100                                            |              |

## Software baseline

Ubuntu 24.04.2 LTS, NVIDIA driver 570.211.01, CUDA 12.8, Python 3.12 venv with PyTorch 2.10+cu128. Tailscale for remote access. Details in ATLAS\_OPERATIONS.md.

## Rough used-market pricing (September 2026, unverified)

Only items with a price found on a live listing are quoted. Check current listings before buying; these move week to week.

| Item                                | Observed price                                                                                                      | Source                             |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Quadro GV100 32 GB, refurbished     | about \$8,000 each from a refurbisher; eBay used listings are typically far lower but were not priced in the search | IT Creations listing               |
| HGST He8 8 TB SAS, used             | about \$45 to \$90 each                                                                                             | eBay sold listing, eBay.de listing |
| HP Z840 barebone with 2x E5-2690 v3 | listings exist on eBay, Newegg Refurbished and PCSP; prices not shown in search results                             | see links in the notes below       |

## Notes for a student build

- The Z840 platform itself (chassis, board, PSU, two E5-2690 v3, 128 to 256 GB RDIMM) is cheap and plentiful used. The GV100s are the expensive and hard part.
- A GV100 is a 2018 Volta card. For the same money a used RTX A6000 (48 GB) or two used RTX 3090 (24 GB each) gives more VRAM per dollar and newer CUDA features, and both fit the Z840 with the 1125 W PSU. Verify card length and the 8-pin power leads before buying.
- The Z840 has no NVLink path between CPUs, so plan on single-GPU training or data-parallel jobs, not model-parallel jobs that depend on fast peer access.
- Thermals are the real limit. Air or liquid, the CPUs hit 99 °C under 48-thread load. Budget for the 1125 W PSU and expect to throttle CPU jobs.
- A PCIe M.2 adapter is needed for the NVMe boot drive; the Z840 has no M.2 slot on the board. Boot from NVMe works with BIOS M60 v02.34.